

# Virtual Reality for inward contemplation

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## ABSTRACT

When performed at a high level, meditation and contemplative practices can give rise to altered states of consciousness, the study of which is key for cognitive neurosciences. Those effects indeed can help to better understand the underlying mechanisms of the self and its relation with the body. But reaching those states is rare and thus hard to observe. They require intense practice and dedication, even among regular meditators. To make those specific states more accessible, we propose a new approach of a “neuro-engineered meditation”, bringing technologies such as virtual reality (VR) and biofeedback to give practitioners new perspectives over their practice, by targeting their own body and internal sensations. We expect this approach to help participants better understand complex concepts and develop deeper results, in comparison with a regular practice.

## CCS CONCEPTS

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; • **Applied computing** → *Health informatics*; • **Computing methodologies** → **Mixed / augmented reality**.

## KEYWORDS

Virtual-reality, contemplative practices, self-consciousness, biofeedback

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## 1 INTRODUCTION

Meditation is well known for bringing positive outcomes to practitioners—such as raising self-awareness, decreasing stress, and improving focus—but it can also be a way to access altered states of consciousness, resulting in various sorts of distancing from the body, such as non-dual awareness, no-self, or transcendental meditation. Within the field of cognitive neuroscience, those states are yet to be understood and remain fascinating open-questions.

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Study of self-consciousness in cognitive neuroscience faces difficulties for experimentally inducing specific states, and for ensuring that experimental manipulations actually leads to the expected changes. If meditation can cultivate reaching different states of consciousness, then merging its approaches to experimental research could overcome some of these limitations. To this matter, we present here a set of tools dedicated to help novice and regular practitioners in exploring and developing their practice of meditation, thanks to virtual reality technology and biofeedback.

## 2 VR FOR INWARD CONTEMPLATION

VR has previously been used for studying and enhancing meditation practice. In the project Dynecom, Jarvela et al. (2019) studied interpersonal synchrony thanks to real-time visualisation of dyadic body signals. Van Rooij et al. (2016) proposed a meditative experience where the pace of the scenario is controlled by the user’s breath. In ZenVR, Feinberg et al. (2022) focused on the storytelling, and the capability of VR to immerse users into a context that fosters their engagement into the task, like a zen style classroom. VR can also bring gamification tools to the meditation practice to make it more playful [see TRIPP or Playne].

These examples show different approaches of what VR can bring to meditation. Building on previous works on bodily self-consciousness, it has been shown that the underlying mechanisms of what permits someone to experience having a ‘self’ are deeply linked with the capacity to inhabit a body [Blanke and Metzinger 2009]. Thus, we consider that it is through its ability to represent and put in perspective a body that meditation in VR can reach its most interesting results. This hypothesis is in line with Wang et al. (2022) who, in their recent review, rightfully proposed that further research should put more emphasis on self-representation.

## 3 OUR APPROACH

To help direct the focus inward, we propose here to provide our participants a set of specific verbal guidance, complemented with a visual shift of perspective over their own body and internal sensations.

Thanks to an in-house volumetric reconstruction tool [Albert et al. 2019], we scan in real time the users’ bodies and integrate them into the virtual reality scene. Then, by placing the virtual camera between the eyes of the reconstructed body, we can give our participants the impression of being embodied into their real body, at a normal first-person point of view (Fig.1). Then, by simply changing the virtual camera point of view, we can shift their perspective to any other location, while the reconstructed body stays in place (Fig.2).

This virtual simulation of an out-of-body experience—a form of alteration of self-consciousness known to be a life-changing



**Figure 1: 1st person point of view of the scene**



**Figure 2: out-of-body perspective over the user's body**

experience—is then associated with specific verbal guidance to bring participants into specific meditative states.

Drawing inspiration from traditional meditation practices, we can then develop several new neuro-engineered meditation techniques:

**Being an observer of oneself** A common exercise in meditation consists in observing oneself being traversed by random thoughts and emotions, and to let them pass. By shifting our participant's points of view to this out-of-body location, we complement the self-observation exercise with a real distancing from the body. Participants see themselves while practising, leading to stronger senses of disembodiment.

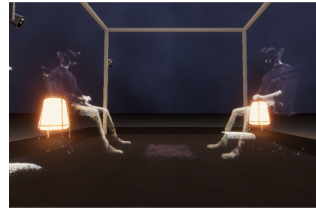
**Focus on one's internal sensations** Internal awareness practices can involve a prolonged focus on the breath or the heartbeat. The ability of our system to represent one's body from an outside perspective can bring the possibility to facilitate this specific exercise: by equipping our participants with ECG sensors and breathing belts, we can stream their body signals and represent their fluctuation onto the volumetric body. In our case we chose to make the body luminance react in synchrony with the input signal, linking the focus on internal sensations with visual cues.

But more than just being a helper for focus, it has been shown that this kind of visualisation can strengthen the sense of body ownership [Adler et al. 2014]. More interestingly, by shifting from a synchronous visualisation to an asynchronous one, where biofeedback are delayed, we can voluntarily break this link and provoke an increased distancing from the body. We believe that this kind of intervention, when combined with appropriate guidance, can lead to even stronger results.

**Self-compassion meditation** Self-compassion meditation consists in sending love to oneself and to one's body. This specific practice can be complex for novices, especially the ones with a low self-esteem. Here, we place our participants in front of a shifted and rotated visualisation of themselves, which sits in their direct sight (Fig.3 & Fig.4). Seeing the body and projecting thoughts on it makes the exercise easier. Specific parts of the body can then be kept into focus for longer periods.

## 4 PRELIMINARY RESULTS

In the three studies we conducted so far, participants reported a stronger effect and a general better quality of experience while being presented this shifted point of view, in comparison to regular



**Figure 3: The user is facing his own avatar**



**Figure 4: 1st person point of view of the facing avatar**

practices without any change of perspective. Over the population of our participants, who were all novices in meditation, a vast majority did enjoy the practice overall and reported being interested in doing the same exercise again in the future. These preliminary results seem to validate our approach and the fact that a visualisation of the body during a task of internal focus can help novices to grasp complex concepts and reach deeper meditative states within a shorter period of time.

## 5 CONCLUSION & FUTURE DIRECTIONS

We proposed here several approaches to go beyond traditional contemplative techniques, by directly involving a visual representation of the body in virtual reality. Along with specifically designed audio sequences and real time feedback, this body appearance can be shifted and altered, to strengthen and reach the specific effects we target. Beyond the validation of our different scientific hypotheses, we expect as an outcome the rising of new types of techniques to learn meditation, aided by neuroscience, and immersive technology. Neuro-engineered meditation could then lead to deeper long-lasting effects, making meditation more accessible, and beneficial to an even wider audience.

## ACKNOWLEDGMENTS

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